

ANANYA SRIDHAR

AI/ML Engineer — Generative AI, LLMs & Agentic Systems

Mumbai, India · +91 8104921962 · ananyasridhar108@gmail.com

github.com/darayavahoush · darayavahoush.github.io · vaaksudhi.manaslearning.com · vaakify.manaslearning.com · linkedin.com/in/ananya-sridhar-ai

PROFESSIONAL SUMMARY

Full-Stack AI/ML Engineer specializing in production-grade LLM architectures, multi-agent workflows (LangGraph), and deterministic RAG systems. Track record of taking complex machine learning architectures—from multi-agent coordination and reinforcement learning to acoustic speech verification and vision transformers—into low-latency, scalable production software. Experienced in high-throughput FastAPI backends, vector search, and edge deployment across healthcare, legal, and fintech domains.

TECHNICAL SKILLS

- **AI & Deep Learning:** PyTorch, Hugging Face (Transformers, PEFT/LoRA, Accelerate), LangGraph, LangChain, ChromaDB, Dense RAG, Reinforcement Learning (Q-learning, DRL), MCTS, Scikit-Learn
- **NLP & Speech AI:** Whisper, Faster-Whisper, Vosk, Coqui TTS, Kokoro ONNX, Librosa, Sarvam AI, DistilBERT, BART, Acoustic Signal Processing, Phoneme Verification, Epitran G2P
- **Computer Vision & Robotics:** OpenCV, MediaPipe Tasks, DINOv2 (Vision Transformers), Optimal Transport (POT), MuJoCo 3D Physics, Extended Information Filter (EIF)
- **Backend & Systems:** Python, C++, SQL, FastAPI, Uvicorn, PostgreSQL, Supabase, Azure Cloud, Docker, WebSockets, REST APIs, Git, Linux, React, Vite

PROFESSIONAL EXPERIENCE

Manas Learning — AI/ML Engineer Intern (Full-Stack & Speech AI)

Mar 2026 — Present | Remote / Mumbai, India

- **LangGraph Diagnostic Pipeline:** Architected an autonomous agentic assessment workflow chaining Faster-Whisper, Vosk, and Kokoro TTS with phoneme-level acoustic verification (Librosa, Epitran) across 8 Indian languages.
- **Adaptive Reinforcement Learning:** Formulated a Q-learning difficulty controller dynamically tuning clinical assessment exercises in real time based on cumulative pronunciation accuracy and response latency.
- **Full-Stack Architecture & Security:** Engineered high-performance FastAPI backends, PostgreSQL schemas, and React + Vite therapist dashboards with COPPA-compliant parental consent flows.
- **Real-World Impact:** Shipped production platforms vaaksudhi.manaslearning.com and vaakify.manaslearning.com with sub-300ms audio latency, enabling automated articulatory error analysis across 8 regional Indian languages.

Nivesh AI — AI/ML Engineer Intern (Enterprise AI & RAG)

Jun 2026 — Present | Mumbai, India

- **Enterprise RAG Pipeline:** Developed a zero-hallucination summarization engine on Azure processing advisor-client audio and portfolio documents for 10,000+ mutual fund distributors (MFDs).
- **Speech-to-Text Optimization:** Evaluated and fine-tuned multilingual STT models (Sarvam AI, Whisper) with spectral denoising (`noisereduce`), boosting transcription fidelity on mixed Indian English/Hindi calls.
- **Structured Data Extraction:** Automated ingestion of complex PDF/DOCX financial portfolios into vector stores using `pdfplumber` and embedding models.
- **Real-World Impact:** Reduced financial advisors' manual documentation and portfolio summarization turnaround time by 80% across 10,000+ active distributor accounts.

FEATURED PROJECTS

Vaaksudhi — Pediatric Speech & Diagnostic Multi-Agent Platform

[Live Platform](#) · [GitHub: Agentic-AI](#)

Python, LangGraph, Faster-Whisper, Vosk, Kokoro ONNX, Librosa, Q-Learning, FastAPI, React

- Architected an autonomous multi-agent clinical diagnostic pipeline coordinating 4 clinical assessment modules (BreathQuest, VaakMirror, Chime, VoiceHurdleRace) across 8 Indian languages.
- Implemented LangGraph state machines evaluating place, manner, and voicing errors with sub-300ms phoneme alignment, auto-generating structured diagnostic clinical reports.
- **Real-World Impact:** Live clinical tool serving pediatric speech assessments in 8 Indian languages with automated clinical report generation.

Vaakify — Pediatric Multilingual Speech Practice Platform

[Live Platform](#) · [GitHub: Speech-Repeater](#)

Python, Librosa, Praat-Parselmouth, Faster-Whisper, Coqui TTS, FastAPI, React

- Built an interactive speech practice platform featuring animated companions and real-time phoneme biofeedback for children with speech and communication challenges.
- Extracted acoustic biomarkers (formant frequencies F1/F2, jitter, shimmer, pitch contours) to identify articulatory drift and dynamically construct personalized corrective drills.
- **Real-World Impact:** Deployed clinical speech companion actively used by children and therapists for gamified phoneme therapy in English, Hindi, and Kannada.

NIRVANA (Firebot) — Autonomous Firefighting Robot & MuJoCo Simulation

[Live Render Console](#) · [GitHub: Firebot](#)

Python, MuJoCo 3D Physics, PyTorch, Gymnasium, EIF Sensor Fusion, FastAPI, WebSockets

- Engineered an autonomous firefighting robot in a physics-backed MuJoCo 3D simulator across procedural multi-room environments with volumetric particle fire modeling.
- Implemented MM-FusionRL (multimodal cross-attention deep RL) coupled with Extended Information Filter (EIF) sensor fusion for thermal localization and obstacle navigation.
- **Real-World Impact:** Deployed live simulation console on Render eliminating hazardous human entry in structural fires by localizing heat sources in <45s inside physics environments.

Tripwire — Real-Time CPU-Only Edge CV Vehicle Counter

[Live Demo](#) · [GitHub: Vehicle-Counter](#)

Python, OpenCV, FastAPI, React, Vite, WebSockets, Docker

- Engineered an edge-ready, GPU-free computer vision vehicle detection and tracking pipeline from live CCTV feeds operating with sub-50ms latency.
- Implemented Gaussian background subtraction, morphological noise filtering, centroid tracking, and directional virtual tripwire counting streamed over WebSockets.
- **Real-World Impact:** Cuts camera surveillance hardware costs by 80% by delivering accurate multi-lane traffic tracking on standard commodity CPUs without dedicated GPUs.

GECO-Enhanced — Confidence-Aware Semantic Keypoint Correspondence

[GitHub: GECO](#)

PyTorch, DINOv2 (ViT), Python Optimal Transport (POT), FastAPI, React, Docker

- Formulated an unsupervised semantic correspondence framework establishing keypoints across different instances of identical object classes under severe pose and perspective variation.
- Extracted dense DINOv2 ViT features and solved correspondence using Sinkhorn-regularized Optimal Transport with confidence weighting and entropy regularization.
- **Real-World Impact:** Enables zero-shot visual alignment across varying camera viewpoints without requiring expensive annotated paired training datasets.

EDUCATION & CERTIFICATIONS

- **Bachelor of Technology (B.Tech),** SASTRA University (Expected 2027)
- **SWAYAM – IIT Madras:** Comprehensive Electronics and Embedded Systems (*In-Person Practical Program*)
- **IIoT Certification:** Ministry of Heavy Industries (MHI) Program (*Project-Based Evaluation*)